

Prediction of Alzheimer’s disease progression from longitudinal blood DNA methylation

Abstract

Two binary classification tasks were evaluated using pre-processed ADNI blood DNA methylation matrices: Control→Control versus Control→MCI, and MCI→MCI versus MCI→Alzheimer’s disease/dementia. Each sample contained two timepoints and 2,000 CpG features selected by variance ratio. A sequence of models was tested, beginning with regularised logistic regression and extending to XGBoost, support vector machines, supervised latent-space classifiers, multilayer perceptrons, Siamese timepoint networks, genome-ordered CNNs, and a chromosome-branch CNN. Across both tasks, regularised linear models gave the strongest and most stable cross-validated performance. Deep learning models provided useful architectural comparisons but did not improve on simpler baselines, consistent with the small-sample, high-dimensional setting.

Data and exploratory analysis

The supplied HDF5 file contained methylation tensors with shape CpGs $\times$ timepoints $\times$ samples. For modelling, arrays were transposed to samples $\times$ timepoints $\times$ CpGs. Task 1 comprised 190 subjects (147 Control→Control, 43 Control→MCI), and Task 2 comprised 301 subjects (191 MCI→MCI, 110 MCI→AD/dementia). Each subject contributed two methylation profiles over 2,000 pre-selected CpG probes. Feature views included t_0 , t_1 , $\Delta = t_1 - t_0$, and concatenated versions of these views.

Initial exploratory analysis described the HDF5 dimensions and visualised the top 200 variable CpGs at each timepoint. These heatmaps showed structured correlation and sample clustering, but they were treated as unsupervised exploratory support rather than evidence of predictive performance. This distinction was important because supervised CpG ranking outside cross-validation would risk information leakage in a small n , high p setting.

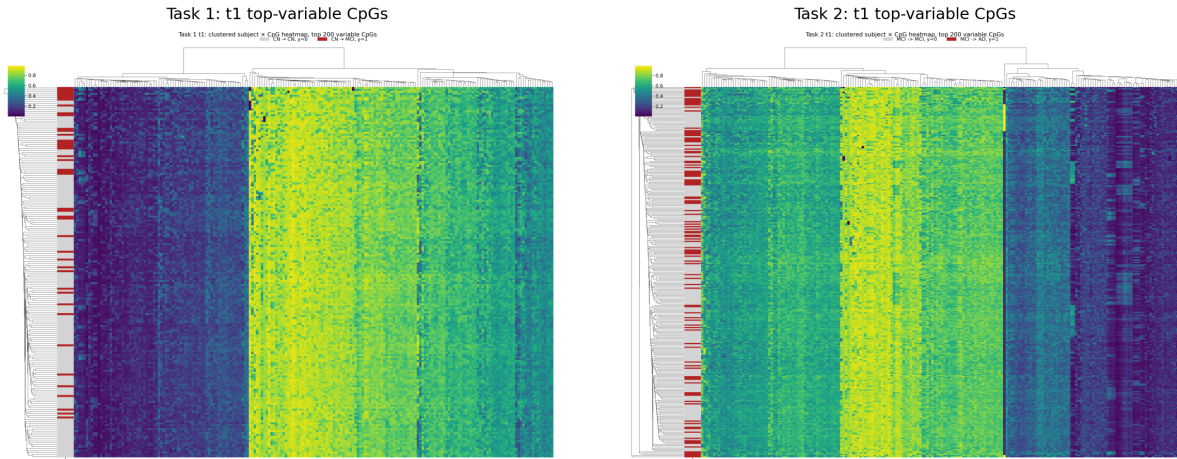


Figure 1: Unsupervised clustermaps of the top-variable t_1 CpGs for Task 1 and Task 2. These plots show methylation structure but were not used as validation evidence.

Modelling strategy

The modelling sequence was deliberately conservative. Recent blood methylation studies in cognitive impairment show promising but variable performance, with external validation often weaker than internal cross-validation; the literature also supports regularised linear and latent-space models as strong first-line choices for methylation data [1, 2, 3]. Therefore, simple regression baselines were fitted before more expressive models.

Baseline models included majority class prediction, L2 logistic regression, elastic-net logistic regression, and PCA-logistic regression. A second classical pass tested linear SVMs, ridge-style classifiers, SGD linear models, PLS-logistic regression, and

PCA/RBF SVMs. XGBoost was evaluated on raw, PCA, and selected feature representations, including a threshold-tuned F1 analysis; threshold-tuned XGBoost results are useful for operating-point exploration but are less directly comparable with default-threshold baselines.

Deep learning experiments used PyTorch with fold-wise standardisation, class-weighted binary cross-entropy, dropout, weight decay, and early stopping. Architectures included regularised MLPs, bottleneck MLPs, Siamese timepoint MLPs using shared encoders for t_0 and t_1 , genome-ordered 1D CNNs, and a chromosome-branch CNN. Recurrent models were not prioritised because two timepoints provide little sequence length for an RNN/LSTM inductive bias; temporal structure was instead represented through paired encoders and explicit Δ features. Model evaluation used repeated stratified cross-validation with ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, and F1.

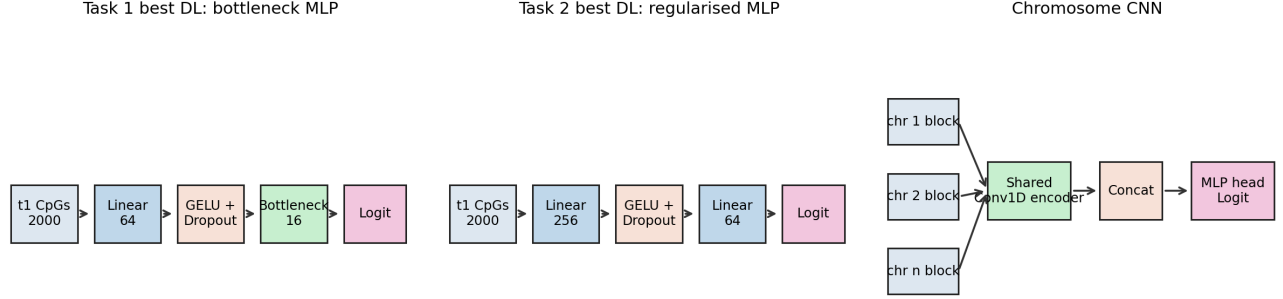


Figure 2: Deep architectures evaluated after the classical baselines. The chromosome CNN applies a shared 1D convolutional encoder within genome-ordered chromosome blocks, concatenates chromosome embeddings, and classifies with a small MLP head.

Results

Figures 3 and 4 summarise representative high-performing models from each family. For Task 1, L2 logistic regression on t_1 achieved mean ROC-AUC 0.941, PR-AUC 0.901, balanced accuracy 0.871, and F1 0.759. The best F1 was obtained by a linear SVC with stronger regularisation on t_0 (F1 0.773), while ridge and PLS variants gave the strongest ranking metrics (ROC-AUC up to approximately 0.954). XGBoost performed best after PCA and threshold tuning, but did not clearly exceed the linear baselines. The best Task 1 deep model was a bottleneck MLP on t_1 (ROC-AUC 0.868, F1 0.693), below the linear models.

For Task 2, L2 logistic regression on the concatenated $t_0 + t_1$ view was the best overall model, with mean ROC-AUC 0.903, PR-AUC 0.878, balanced accuracy 0.826, and F1 0.776. PLS-logistic regression on t_0 was competitive (F1 0.744). XGBoost again lagged the classical linear/latent-space models. The strongest Task 2 deep model was a regularised MLP on t_1 (ROC-AUC 0.829, F1 0.701), which improved over the CNNs but remained below logistic regression.

Task	Model	ROC-AUC	PR-AUC	Bal. acc.	F1
1	Logistic L2, t_1	0.941	0.901	0.871	0.759
1	Linear SVC, t_0	0.924	0.864	0.876	0.773
1	Best DL bottleneck MLP, t_1	0.868	0.787	0.810	0.693
2	Logistic L2, $t_0 + t_1$	0.903	0.878	0.826	0.776
2	PLS-logistic, t_0	0.877	0.854	0.800	0.744
2	Best DL MLP, t_1	0.829	0.782	0.767	0.701

Table 1: Selected cross-validated mean metrics for the strongest representative models.

Discussion

The dominant pattern was that simple, regularised decision boundaries generalised better than more expressive nonlinear or deep architectures. This is plausible for methylation data with far fewer samples than features, correlated CpGs, and pre-selected probes that may already enrich for approximately linearly separable signal. Elastic-net logistic regression was useful as a sparsity check,

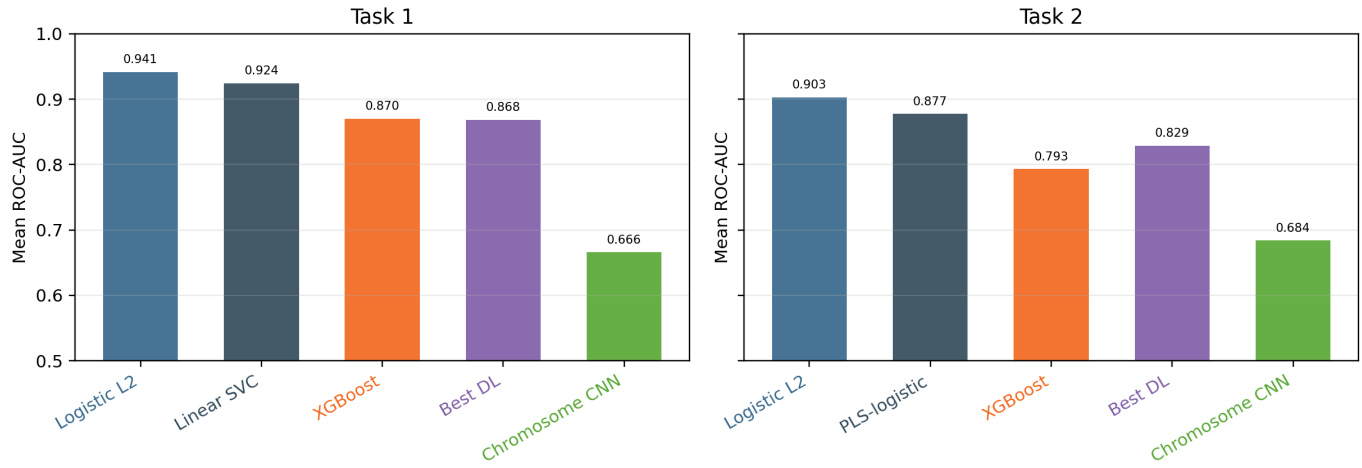


Figure 3: Cross-validated mean ROC-AUC for representative model families. Classical regularised models performed best on both tasks; CNN variants were weaker despite biologically motivated ordering.

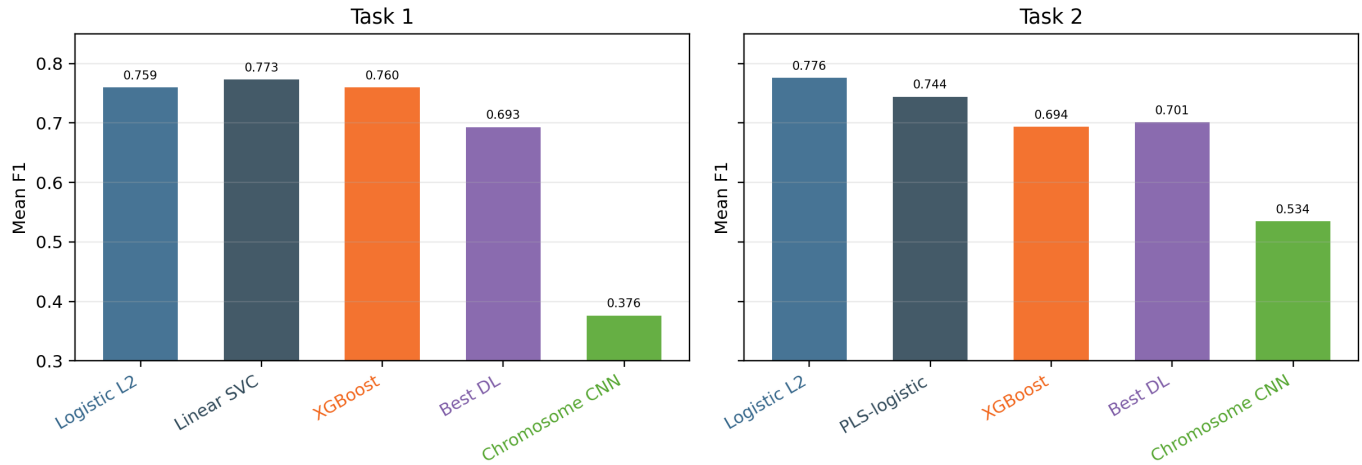


Figure 4: Cross-validated mean F1 for representative model families. The ranking is broadly similar to ROC-AUC, but threshold-sensitive models such as linear SVC and XGBoost move slightly relative to logistic regression.

but it did not improve on L2 logistic regression, suggesting that strongly sparse feature selection was not essential for the best internal performance. PLS-logistic and ridge/SVM models were also strong, consistent with a distributed methylation signature.

The deep learning experiments were informative despite not winning. Small MLPs performed better than CNNs and Siamese models, especially on t_1 , indicating that the later timepoint carried most of the usable signal. Genome-ordered CNNs and the chromosome-branch CNN did not improve performance, suggesting that local genomic adjacency was not a strong enough inductive bias for this 2,000-CpG subset. A nested-CV optimisation pass re-selected the same tabular MLP family and did not improve on earlier tuned-small runs, supporting the conclusion that generalisation, not model capacity, was the limiting factor.

Important limitations remain. Evaluation was internal cross-validation only, with no independent external cohort. Metadata on age, sex, blood cell composition, APOE, medication, assay batch, and other confounders was not incorporated; these factors can strongly influence blood methylation and may inflate apparent disease signal if imbalanced. With more time, the next extensions would be fold-safe supervised CpG selection or PCA inside cross-validation, and a joint multi-task encoder sharing representation learning across both progression tasks. External validation and calibration analysis would be necessary before any clinical interpretation.

References

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